recognition skills which again is a blend of various technologies and spans multiple research streams.

Recognition based AR

Recognition based AR depends heavily on pattern recognitions. More your app knows about patterns, more it can do. Pattern recognition technology has developed in leaps and bounds in the last half decade. Smile shot, face recognition for tagging and Google's 'Search by image' feature are some of the most famous applications of image recognition technology. But this is nowhere close to completion.

The first place from where you can start off with recognition based AR development is, once again, your smartphone because it already comes with the features of pattern recognition. Every time you scan a QR code to install an app, to open a webpage, you employ the pattern recognition faculties of your phone. But your phone can only get good enough for basic pattern recognition. If you want it to do advanced 3D object recognition, it might not prove to be the best gadget.

QR code scanning depends on pattern recognition

3D object recognition requires much more data, processing power and efficient algorithms for getting the job done. While data can be stored on a phone and algorithms be designed for the same, phones cannot have very powerful processors. Coming back to 3D object recognition, you would typically be in need of clever image recognition systems which know how an object appears from different angles.

If you have to create an AR application which can recognise objects and then augment a layer on top of it, you are limited by some very basic shapes. For example, a popular development library AndAR (stands for Android Augmented Reality) which allows you to create recognition based AR applications for the Android platform supports only very basic pattern recognitions. Better computer vision and object recognition libraries are available for more powerful systems (such as for desktops and laptops) and with their aid, you can create more intelligent applications. It is easily predictable that in time, with more advanced hardware and software, such libraries will be available for mobiles as well.